Ablation Study Results

Overview

We systematically ablate individual model components to identify which mechanisms are necessary for the emergence of rhythmicity in Group 2 (the non-rhythmic group). Each condition was run with 5 independent seeds and 200 permutations for the rhythmicity test. We report the sine-fit amplitude A and the permutation p-value for Group 2.

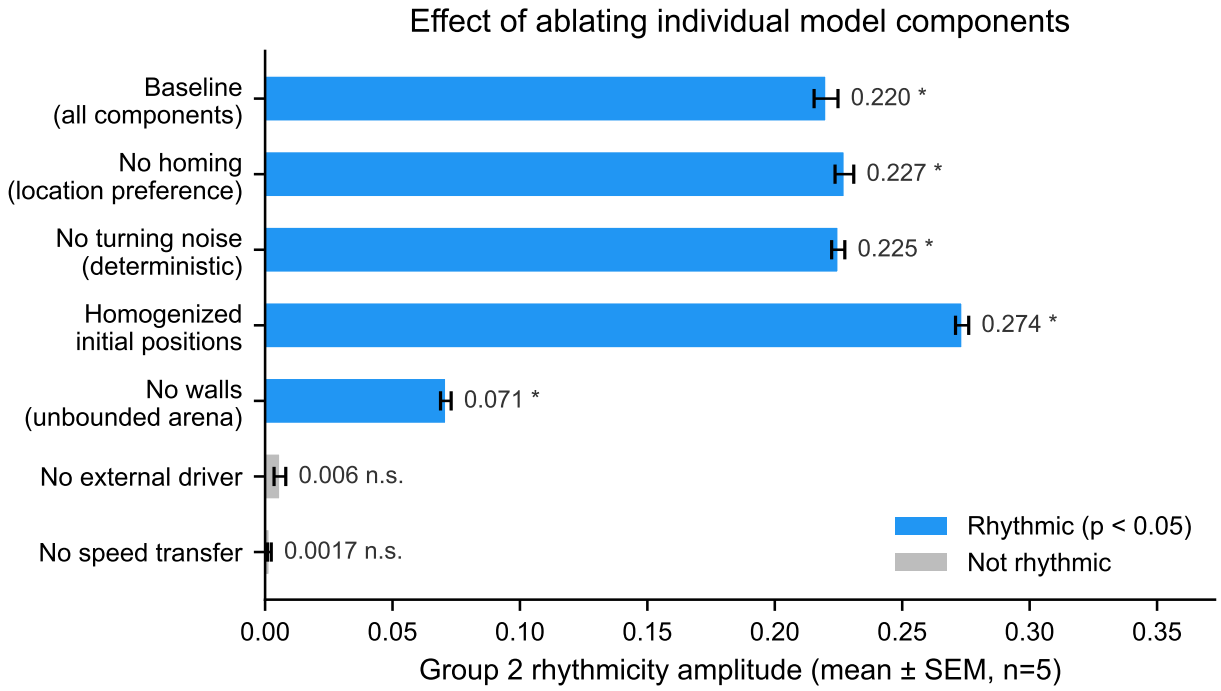


Figure 1: Effect of ablating individual model components on Group 2 rhythmicity amplitude (mean ± SEM, n=5 seeds). Speed transfer is the only component whose removal eliminates Group 2 rhythmicity entirely ($A_2 \approx 0.002$, $p = 0.36$). Walls have a secondary amplifying effect, reducing A_2 from 0.223 to 0.059 (confinement increases contact rate). Homing and turning noise have negligible effects. Homogenizing initial positions slightly *increases* transfer ($A_2 = 0.234$), since mixing groups spatially produces more early inter-group contacts.

Supplementary: Parameter Sensitivity

To assess the robustness of the speed-transfer model, we performed single-parameter sweeps over the three main tunable parameters: agent density d, day duration T_{day}, and decay rate γ. In each sweep

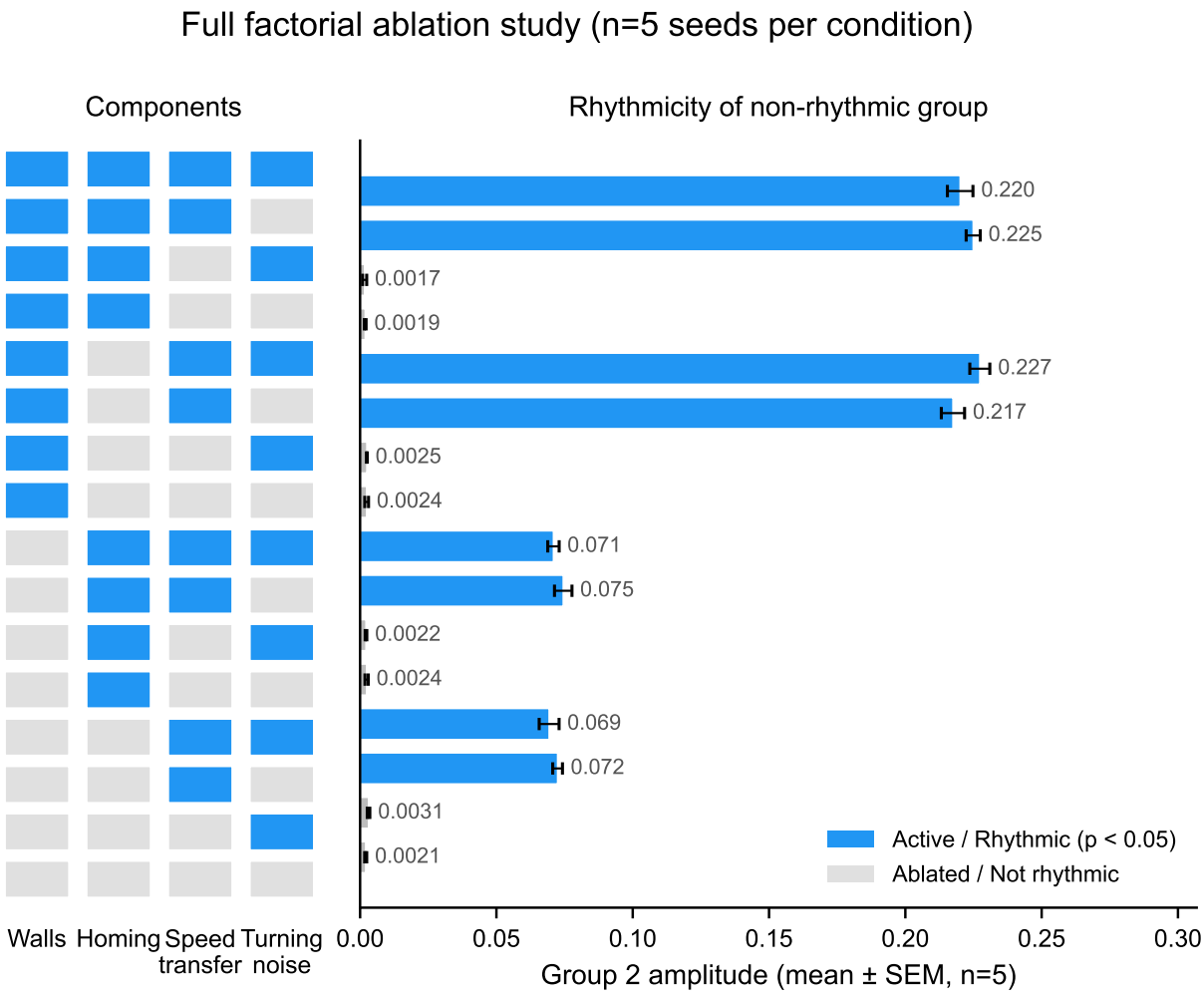


Figure 2: Full 2^4 factorial ablation over four structural components (n=5 seeds per condition). Left: blue squares indicate the component is active; grey indicates it has been ablated. Right: Group 2 amplitude. Whenever speed transfer is ablated (grey in the third column), Group 2 amplitude collapses to near zero regardless of other component states.

one parameter is varied while all others are held at their baseline values (d=1.0, T_{day}=100 steps, γ=0.8); each condition is evaluated with n=5 independent seeds and 200 permutations.

Summary of parameter effects

Density (d). The density factor scales the number of agents in the arena and thereby controls the availability of interaction partners. Higher density increases the rate of inter-group contacts, which leads to faster build-up of rhythmicity in Group 2 (larger A_2) and correspondingly shorter propagation delays (smaller $\Delta\phi$). Both effects are monotonic: A_2 increases approximately linearly from 0.07 at d=0.25 to 0.37 at d=3.0, while $\Delta\phi$ decreases from $\sim$17° to $\sim$9° (Fig. 3).

Day duration (T_{day}). The day duration sets the number of simulation steps per light–dark cycle. A longer cycle provides more samples per day, making the temporal resolution finer, but it

Table 1: Ablation study: effect of removing individual model components on rhythmicity transfer (mean ± SEM across n=5 independent seeds). A_1 and A_2 are the fitted sinusoidal amplitudes of the Group 1 (forager) and Group 2 (hive bee) mean speed time series, respectively. $\Delta\phi$ is the phase lag of Group 2 relative to Group 1 in degrees, and p_2 is the permutation-test p-value for Group 2 rhythmicity (200 permutations). Simulations used density factor d=1.0, day duration T_{day}=100 steps, and decay rate γ=0.8. Speed transfer is the sole essential mechanism; walls provide secondary amplification by confining agents and increasing contact rates.[a]

Condition	A_1	A_2	$\Delta\phi$ (°)	p_2
Baseline (all components)	1.25 ± 0.00	0.220 ± 0.005	14.8 ± 0.6	<0.01
No speed transfer	1.25 ± 0.00	0.0017 ± 0.0008	—	0.65
No homing (location preference)	1.25 ± 0.00	0.227 ± 0.004	12.7 ± 1.3	<0.01
No walls (unbounded arena)	1.25 ± 0.00	0.071 ± 0.002	20.2 ± 4.3	<0.01
No turning noise	1.25 ± 0.00	0.225 ± 0.003	14.8 ± 0.6	<0.01
No external driver	0.01 ± 0.00	0.006 ± 0.002	—	0.36
Homogenized initial positions	1.24 ± 0.00	0.274 ± 0.003	8.6 ± 1.0	<0.01
No homing + no speed transfer	1.25 ± 0.00	0.0025 ± 0.0002	—	0.31
No walls + no speed transfer	1.25 ± 0.00	0.0022 ± 0.0003	—	0.42
No walls + no homing	1.25 ± 0.00	0.069 ± 0.004	7.7 ± 1.6	<0.01
No walls, homing, or speed transfer	1.25 ± 0.00	0.0031 ± 0.0005	—	0.21

[a]Phase shift omitted where Group 2 rhythmicity was not significant ($p \geq 0.05$).

also means the exponential decay γ is applied more times per day—the effective per-day memory is $\gamma^{T_{\text{day}}}$, which becomes negligibly small for large T_{day} (e.g. $0.8^{400} \approx 10^{-39}$). In practice, amplitude is largely invariant to day duration ($A_2 \approx 0.22$ for $T_{\text{day}} \geq 75$), confirming that the steady-state transfer signal does not depend on temporal resolution. By contrast, phase lag scales inversely with T_{day}: $\Delta\phi \approx 31°$ at T_{day}=50 down to $\Delta\phi \approx 2°$ at T_{day}=800 (Fig. 4). This relationship arises because the propagation delay is approximately constant in absolute time steps; as the cycle length increases, this fixed delay represents a progressively smaller fraction of the full 360° cycle.

Decay rate (γ). The decay rate implements a leaky memory of past speed transfers: at each time step, the accumulated speed-transfer signal is multiplied by γ, so γ=0 corresponds to no memory (only instantaneous interactions) and values approaching 1 retain nearly all past information. Amplitude increases monotonically with γ ($A_2 \approx 0.06$ at γ=0 to $A_2 \approx 0.30$ at γ=0.9), because longer memory allows the rhythmic signal to accumulate across multiple interactions (Fig. 5). Crucially, rhythmicity transfer remains significant ($p < 0.01$) even at γ=0, demonstrating that the basic mechanism does not require temporal integration. Phase lag also increases substantially with γ, approximately tripling from $\Delta\phi \approx 7°$ at γ=0 to $\Delta\phi \approx 21°$ at γ=0.9. This is because the accumulated memory acts as inertia: at high γ, an agent's speed is heavily influenced by interactions from many previous steps, so it takes longer for the current phase of the driving rhythm to override the historical signal, delaying the response of Group 2.

Figures

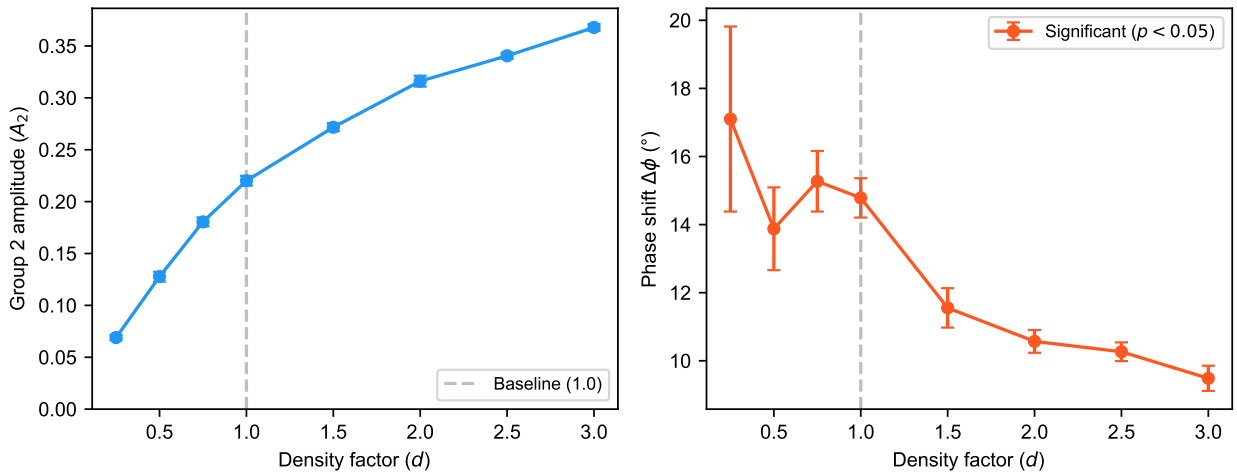


Figure 3: Sensitivity to agent density ($d \in [0.25, 3.0]$, T_{day}=100, γ=0.8). **Left:** Group 2 amplitude A_2 increases monotonically with density. **Right:** Phase lag $\Delta\phi$ decreases with density, as more available interaction partners accelerate signal propagation. Dashed line marks the baseline (d=1.0).

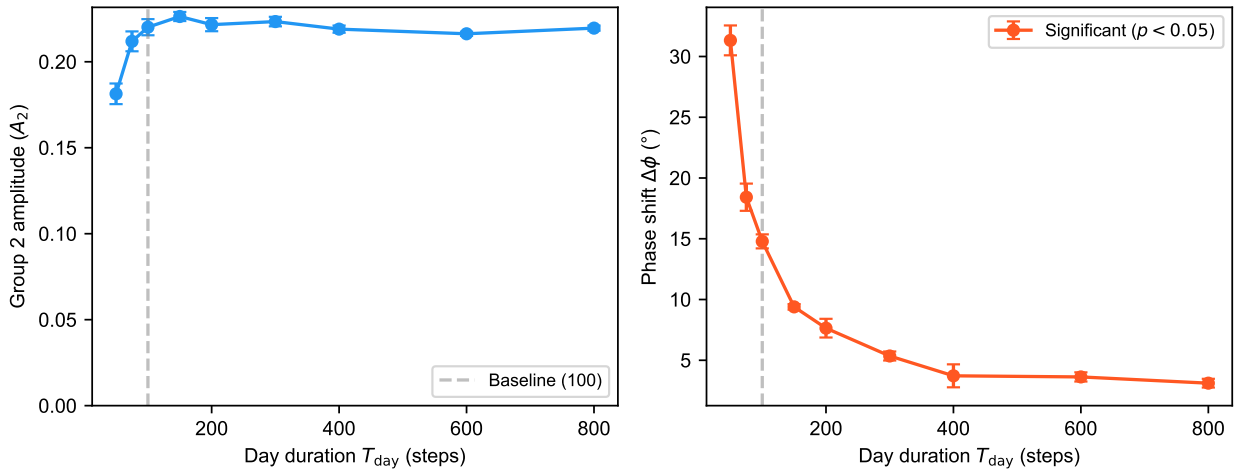


Figure 4: Sensitivity to day duration ($T_{\text{day}} \in [50, 800]$ steps, d=1.0, γ=0.8). **Left:** Amplitude is nearly constant for $T_{\text{day}} \geq 75$. **Right:** Phase lag scales inversely with day duration, consistent with a fixed propagation delay measured in steps. Dashed line marks the baseline (T_{day}=100).

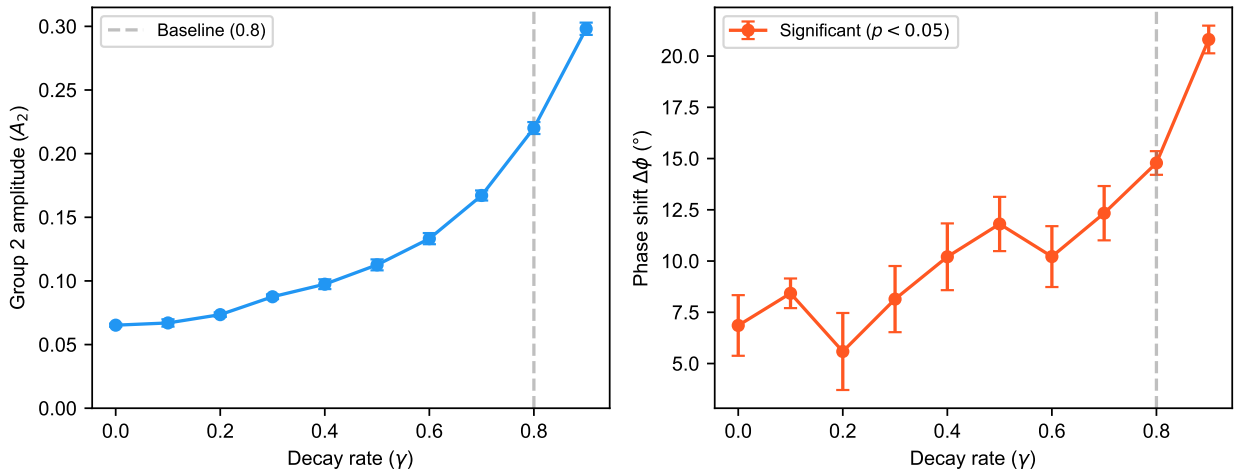


Figure 5: Sensitivity to decay rate ($\gamma \in [0.0, 0.9]$, d=1.0, T_{day}=100). **Left:** Amplitude increases with γ as longer memory amplifies the accumulated signal. Importantly, transfer is significant ($p < 0.01$) even at γ=0. **Right:** Phase lag increases approximately threefold from γ=0 to γ=0.9, reflecting inertia from accumulated memory. Dashed line marks the baseline (γ=0.8).